

Assignment-7

Object Oriented Programming Lab (Java: Polymorphism, Abstract Class, and Interface)

1. You are required to write a base class Student having details (name, registration number, father's name, address (current address, permanent address), and contact details (phone number, email Id). UGStudent, PGStudent, and PhDStudent extend the Student class they have details about the past qualifying degree and marks. Make each attribute protected and provide suitable getter and setter to access attributes. **Use the Java Abstract class concept.**
2. **Create an interface** Vehicle having three methods change Gear, speedup, and apply Brakes. Two other classes, Bike and Bicycle, implement the Vehicle interface. These classes have two fields, i.e., gear and speed. The change Gear method accepts the value of gear and assigns it to the gear field. The speedup method accepts the value by which the speed field should be increased. The apply Brakes method accepts the value by which the speed field should be decreased. Create a Test class to demonstrate each of these classes.
3. Write a class Employee having attributes: employeeID, name, department, dob, designation, yearOfJoining, and phoneNumber. Add a function to print details of employee using employeeID. Two classes, i.e., Faculty and **OfficeStaff**, extends the Employee class. The Faculty has a list of subjects and labs taught by them. **OfficeStaff** has a list of skills which stores skill such as Typing, Technician, etc. Each of the above classes has appropriate getter setter for inserting skill in the list and printing the list of skill. Demonstrate above class using a Test class which creates 5 employees of deferent types and print their details. **Use Java Abstract class concept.**
4. **Create an interface** Shape having two methods getArea and getPerimeter. Three classes, Circle, Triangle, and Rectangle, implement the Shape interface, and **override** the two methods. Create a Test class to demonstrate each of these classes.
5. **Painting Shapes**
Develop a class hierarchy of shapes and write a program that computes the amount of paint needed to paint different objects. The hierarchy will consist of a parent class Shape with three derived classes - Sphere, Rectangle, and Cylinder. For the purposes of this exercise, the only attribute a shape will have is a name and the method of interest will be one that computes the area of the shape (surface area in the case of three-dimensional shapes). Do the following.

Write an abstract class Shape with the following properties: >An instance variable shapeName of type String

>An abstract method area()

>A toString method that returns the name of the shape